CO-PRIOR: Composing Disjoint-Cohort Experts for Flexible Cervical Cancer Screening

Prior-stripped opinion pooling, and the price of an assumption you cannot avoid paying

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Abstract

Cervical cancer is almost entirely preventable, yet the two cheapest screening signals—a Pap-smear cell image and a patient risk questionnaire—are only ever published as *separate* datasets with no shared patients and different label spaces. A flexible screener that accepts either or both is therefore blocked at the first step: every missing-modality method we verified needs a *paired anchor* somewhere (a complete-modality teacher [35], a shared encoder [34], learned priors [24], or paired pretraining). Two failure modes make the naive fix worse than useless. *First*, the tabular arm's published accuracy is an artefact: leakage-safe nested cross-validation puts UCI biopsy prediction at AUROC = 0.656 (95% CI [0.584, 0.720]), against a literature reporting 0.98–1.00 [29]; predicting `Dx:Cancer` from its sibling `Dx` columns alone scores 0.998. *Second*, naive late fusion can *lose*: an image+tabular late-fusion system scored 92% against 96% for its tabular expert alone [19]. We propose **CO-PRIOR**, whose single claim is the *two-prior composition operator*: because each expert's posterior is contaminated by its own cohort's prevalence, we *strip* each prior out before pooling and apply one declared deployment prior afterwards. *Honest novelty note:* the pooling rule itself is **borrowed**—it is ProbEn [5] and, one level up, a weighted opinion pool [10]; calibration across modality subsets is **borrowed** from AECF [6]; only the prior-stripping precondition is **ours**. Crucially it is validatable *per expert* on public data. We evaluate on SIPaKMeD [26], Herlev, HiCervix [3] and UCI [8], against `max(image-only, tabular-only)`, with a pre-registered falsifier.

CCS Concepts: • **Computing methodologies** → **Machine learning**; • **Applied computing** → *Health informatics*.

Keywords: decision-level fusion, missing modality, calibration, prior shift, conformal prediction, cervical cancer screening, data leakage

1 Introduction

1.1 Background for the non-specialist

Cervical cancer is caused by persistent infection with high-risk human papillomavirus (HPV). It progresses slowly through detectable pre-cancerous stages, which is why screening works so well: catch the lesion early and the disease is preventable. The bottleneck is not biology—it is access to the expert who reads the slide.

The two signals this proposal combines

Pap-smear cytology. Cells are scraped from the cervix, stained, and examined under a microscope. A trained cytopathologist looks for abnormal cell shapes. This is a *finding*: it looks at the disease itself.

Risk questionnaire. A form recording age, number of pregnancies, smoking, contraceptive use, and sexually-transmitted-infection history. This is a *pre-test risk*: it estimates how likely a problem is *before* anyone looks.

Two very different colleagues

Think of the image model as a *pathologist* who looks down the microscope at your actual cells, and the questionnaire model as a *receptionist* who glances at your intake form. The pathologist sees the disease; the receptionist only knows that people with your profile tend to be higher risk. You would happily let the receptionist say "bump this patient up the queue"—but you would never let them overrule the pathologist. Almost everything in this proposal follows from taking that asymmetry seriously.

The naive engineering picture is: train a model on each signal, average their outputs, ship a system that works with either. Every step of that picture fails in a way the literature has documented but not solved, and the rest of this section says exactly how.

1.2 The problems

Problem 1: the questionnaire model's reported accuracy is an artefact. The near-universal substrate for cervical risk modelling is the UCI *Cervical Cancer (Risk Factors)* dataset [8]: 858 patients, 36 attributes, four binary screening targets. Published models report 0.98–1.00 AUC on it—for example a stacking ensemble reporting 0.98 accuracy and 0.995

AUC using SMOTE [4] and, by the authors' own statement, no external validation [29].

Why that number is not real

Two mistakes produce it. *(1) Studying the answer sheet.* The dataset ships `Dx:Cancer`, `Dx:CIN`, `Dx:HPV` and `Dx`—records of *past diagnoses*. Predicting one from the others is not diagnosis, it is copying. *(2) Photocopying the exam before splitting it.* SMOTE invents synthetic minority patients; if you do that *before* the train/test split, synthetic copies of a test patient end up in training. The model has seen the test.

This is measured, not hypothetical. Under leakage-safe nested cross-validation—`Dx:*` dropped, resampling confined inside folds—we obtain AUROC = 0.656 (95% CI [0.584, 0.720]) for the biopsy target, and a tuned bake-off over eight model families does not exceed ≈ 0.65. Predicting `Dx:Cancer` from its sibling `Dx` columns scores 0.998; from legitimate features only, 0.592. No single feature exceeds a univariate AUROC of 0.572. The dataset's own creators, using stratified 10-fold with nested inner cross-validation and *no* oversampling, are the honest anchor here [9].

Problem 2: the data is unpaired, and that is structural. No public dataset pairs a cervical image with a risk questionnaire *for the same patient*. SIPaKMeD (4,049 cells from 966 cluster images) [26], Herlev, and HiCervix (40,229 cells from 4,496 slides) [3] carry *no* patient metadata at all. The barrier is consent and de-identification, not effort: the datasets were collected anonymously, so age and history cannot be recovered by re-annotation.

Problem 3: fusion can make things worse, and nobody checks. An image+tabular late-fusion system for cervical risk scored 92%—while its own tabular expert alone scored 96% [19]. Meanwhile calibration, the property that decides whether a screening score can be trusted, is absent: across the missing-modality and cervical-fusion literatures we verified, calibration is not reported at all.

1.3 The gap, and the idea in one picture

Two mature literatures sit either side of our problem and never meet. One *corrects a single classifier's prior* when it is deployed into a population with a different prevalence [1, 11, 31]. The other *combines several classifiers*—but always assuming they share one prior [5, 10, 22].

G3 — the two-prior gap

Nobody performs prior-*stripping* as a **precondition of pooling**. Yet two experts trained on two cohorts necessarily carry *two different* baked-in prevalences: our image expert sees ≈ 40% abnormal cells (a labelling choice), the questionnaire expert ≈ 6.4% biopsy-positive patients (a clinic's case mix). Neither is a deployment prevalence. Pooling them raw counts an accident of dataset construction as if it were evidence.

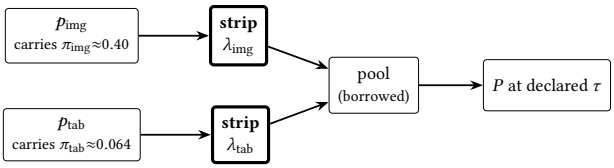


Figure 1. The idea. Each expert's posterior is contaminated by its own cohort's prevalence. We divide that prior out (thick boxes = our contribution), pool the resulting prior-free evidence with a *borrowed* rule, then apply one *declared* deployment prior τ once.

1.4 Contributions

- **The two-prior composition operator (core).** Prior-stripping as a precondition of pooling experts trained on disjoint cohorts—the one operation the prior-correction and classifier-combination literatures each leave to the other. Validatable *per expert*, without any paired cohort.
- **A leakage-audited honest baseline for UCI (core).** No prior work satisfies all three of {`Dx:*` dropped, resampling confined inside folds, calibration reported}. We supply it: 0.656 [0.584, 0.720], with a two-factor decomposition of where 0.98–1.00 comes from.
- **The price of a forced assumption.** At zero cohort overlap the cross-modal dependence is *non-identified*, so conditional independence is forced, not chosen. We transport a quantified, *directional* penalty from biostatistics (over-estimation of risk up to 56% [12]) into a pre-registered prediction about our own output.
- **An honest impossibility result.** We show what a disjoint-cohort screener can and cannot be validated to do—and report the both-modality path as unvalidatable rather than asserting a number.

2 Related Work

2.1 Image-only cervical AI

One line of work classifies single cells with progressively stronger backbones—CNNs, then vision transformers—on SIPaKMeD [26] and Herlev, reporting accuracies above 95%. It works because the benchmarks are curated, balanced, pre-segmented single cells. A second line scales to whole slides with cervical foundation models: Smart-CCS pretrains on 127,471 slides across 48 centres and reports internal AUC 0.965 and external 0.950 [20]. The shared limitation is transfer: our own SIPaKMeD-trained expert scores 0.998 in-distribution

but 0.73 on Herlev, 0.59 on HiCervix and 0.53 on APACC. Worse, SIPaKMeD and Herlev publish *no patient identifiers*, so a patient-wise split is impossible *in principle*—cells cropped from one slide can land on both sides of a random split.

Recurring limitation

These systems classify images ever more accurately on the benchmark they were trained on; *none* reports whether its confidence is trustworthy, and none degrades gracefully when the image is unavailable.

2.2 Tabular cervical risk models

A second family fits classical learners—random forests, boosting, stacking ensembles—to the UCI questionnaire and reports 0.98–1.00 AUC [29]. The pattern is consistent: oversample with SMOTE [4], keep the `Dx:*` columns, evaluate on a single Venezuelan cohort, report accuracy. Gradient-boosted trees are the right model family for tables of this size [13]; the problem is not the learner but the protocol. The honest anchor exists and is by the dataset's own creators, who use nested cross-validation without oversampling [9].

Recurring limitation

No paper satisfies all three of: `Dx:*` excluded, resampling confined inside folds, calibration reported. The field's headline numbers measure leakage, not risk.

2.3 Missing-modality and multimodal fusion

A large literature makes multimodal models survive an absent input. HeMIS-style averaging, SMIL's learned modality priors [24], ShaSpec's shared/specific decomposition [34], MMANet's complete-modality teacher [35], and prompt-conditioned transformers [23] all work—and all need the same thing. As their own survey confirms [36], the mechanism is learned from samples where the modalities are *co-observed*. Even the work that explicitly rejects jointly-pretrained encoders still "requires modality-complete data during training" [21]. Closest to us in spirit: AECF enforces monotone confidence across *all* modality subsets and measures per-subset ECE [6]; Gu et al. fuse image and tabular data with modality dropout [14]. Both train on *paired* data with *simulated* masking.

Recurring limitation

Every method assumes a paired anchor exists somewhere. At *zero* overlap—two cohorts, no shared patient, different label spaces—the anchor cannot be built, and imputation has nothing to impute from.

2.4 Composition, calibration, and abstention

The theory we actually need is older and lives outside cervical AI. Kittler et al. derive the sum and product rules and show the sum rule is more robust to individual estimation error [22]; Genest and Zidek characterise opinion pooling generally [10]; ProbEn applies Bayes' rule with conditional independence and marginalises over absent modalities [5]. On the calibration side, temperature scaling is standard [15], and prior correction has a canonical EM procedure [31] with a modern successor that warns the procedure presupposes calibration—"which is not true of modern neural networks" [1]. Uncertainty can be made explicit with evidential heads [16, 17, 32], though the resulting mass is a *relative* signal, and quality-aware fusion needs the per-view scales to be comparable [37]. Conformal prediction supplies distribution-free guarantees [2], but they are *marginal*, and under label shift they degrade unless unlabeled target data is available [27].

Recurring limitation

One literature retargets *one* classifier to *one* population; the other combines *several* classifiers under a *common* prior. Neither strips priors as a precondition of pooling—which is exactly what two disjoint cohorts force you to do.

3 Research Gaps

G1 — Every missing-modality method needs a paired anchor.

- *Evidence.* Complete-modality teacher [35]; shared encoder trained on co-observed modalities [34]; priors learned where the modality is observed [24]; frozen paired backbone [23]; "requires modality-complete data during training" [21]; confirmed by the field survey [36].
- *Why open.* The dependency is architectural, not incidental. At zero overlap there is nothing to anchor to.
- *This proposal's answer.* M5: an absent expert contributes the additive identity—no imputation, no anchor.

G2 — Calibration is unreported, and nobody asks whether confidence should drop when a modality is missing.

- *Evidence.* Not reported across the missing-modality and cervical-fusion work we verified; the counterexample is AECF, on paired non-medical data [6].
- *Why open.* A screening score that cannot say "I am less sure now" is unsafe in exactly the case flexible inference creates.
- *This proposal's answer.* M3 + H-D: monotone confidence in the evidence set, measured.

Table 1. Gaps mapped to the module that answers them and the question that tests them.

Gap	Module	RQ
G1 paired anchor required	M5 router	RQ4
G2 calibration unreported	M3, M8	RQ4
G3 two priors	**M4 (ours)**	**RQ2**
G4 forced independence	M7 envelope	RQ3
G5 leakage	M2 protocol	RQ1

G3 — No method composes two experts carrying two different priors.

- *Evidence.* Prior correction is one-classifier-one-population [1, 11, 31]; combination assumes a common prior [5, 10, 22].
- *Why open.* It is a hole in the theory, and it is unavoidable once cohorts differ.
- *This proposal's answer.* **M4 — the operator we claim.**

G4 — The only available fusion route is the one that miscalibrates most.

- *Evidence.* Independence over-estimates risk by up to 56% [12]; naive late fusion lost to its own tabular expert, 92% vs 96% [19].
- *Why open.* At zero overlap the dependence is non-identified: the assumption is forced.
- *This proposal's answer.* M7: report an envelope over the unknown dependence, and pre-register the predicted direction of the error.

G5 — Both arms sit on leakage-inflated benchmarks.

- *Evidence.* 0.98–1.00 published [29] vs 0.656 leakage-safe (ours) and the creators' nested-CV anchor [9]; no patient IDs in SIPaKMeD/Herlev.
- *Why open.* Nobody has run the audit on this dataset.
- *This proposal's answer.* RQ1 + ablations A3/A4.

4 Research Questions and Hypotheses

RQ1 (honest baseline). What is the leakage-free ceiling of the UCI questionnaire for biopsy, and how much of the published 0.98–1.00 is leakage? H_0: the leakage-safe AUROC is statistically indistinguishable from the published figure. H_1: it is far lower, and the difference is accounted for by target leakage plus pre-split resampling.

RQ2 (the operator). Does stripping each expert's own prior *before* pooling produce better-calibrated composed posteriors than pooling raw posteriors? H_0: $\Delta\text{ECE} = 0$. H_1: $\Delta\text{ECE} < 0$ (stripping helps), growing with the gap between the two priors.

RQ3 (the price). What is the magnitude and direction of the miscalibration caused by the forced independence assumption, and over what dependence range does composition beat the better single expert? H_0: composition is unbiased and always helps. H_1: it is *over-confident* in a predictable direction, and the gain is ≤ 0 for realistic dependence.

RQ4 (confidence under absence). Is confidence monotone in the available evidence set, and does per-path calibration survive prevalence shift? H_0: confidence is unchanged when a modality is removed. H_1: confidence decreases, and no subset is more confident than a superset containing it.

Pre-registered hypotheses.

H-A. Leakage-safe UCI biopsy AUROC ≤ 0.75 (upper CI bound). *Metric:* AUROC + BCa CI.

H-B. Honest AUROC plus the two leakage increments reconstructs the published figure to within 0.02, leaving no residual signal.

H-C. $\Delta\text{ECE}(\text{stripped} - \text{raw}) < 0$, and its magnitude grows with $|\text{logit}\,\pi_{\text{img}} - \text{logit}\,\pi_{\text{tab}}|$.

H-D. Mean confidence is monotone: both $\geq$ single $\geq$ prior-only, with zero superset violations.

H-E *(falsifier).* *On the realistic operating point, composition does* not *beat* `max(image-only, tabular-only)` *on AUROC beyond noise*—stated up front, because both experts are HPV-driven and therefore positively dependent, which is precisely where pooling stops paying. If H-E holds, we report the negative result; the precedent for it being real is [19].

5 Proposed Methodology

5.1 Design philosophy and honest positioning

CO-PRIOR is deliberately small. Everything that can be borrowed *is* borrowed, so that the contribution is a single, testable operation rather than a system. The two experts are frozen; every component after them is closed-form and costs microseconds; and every component is fitted *on one cohort at a time*, because that is the only regime in which anything here can be validated.

What is NOT novel

We do *not* claim the fusion rule: applying Bayes' rule under conditional independence and marginalising over absent modalities is ProbEn [5], and one level up it is a weighted opinion pool, a literature characterised forty years ago [10]. We do *not* claim calibrated confidence across modality subsets: AECF claims *and measures* it [6]. We do *not* claim modality dropout for image+tabular: that is Gu et al. [14]. We do *not* claim prior correction: that is Saerens et al. [31], superseded by [1]. **Our contribution is the precondition**: stripping each expert's own prior *before* the pool, which neither literature does.

Table 2. Borrowed / modified / new. The novelty is one row.

Component	Status	Closest prior
M1 image expert	borrowed	[26, 33]
M2 tabular expert	borrowed	[13]
M2 leakage protocol	modified	[9]
M3 calibration	borrowed	[1, 15]
M4 prior-stripping	**NEW**	*none found*
M5 availability router	borrowed	[5]
M6 pooling rule	borrowed	[10, 22]
M7 dependence envelope	modified	[12]
M8 conformal abstention	borrowed	[2]
M9 modality attribution	modified	—

5.2 Why not Bethesda?

A reader who knows cytology will object immediately: there *is* a standard shared language for cervical results, so why invent a binary axis?

The Bethesda System (TBS)

TBS is the international standard for reporting cervical cytology [25]. It grades a slide on an ordered scale—NILM (negative), ASC-US (atypical squamous cells of undetermined significance), LSIL (low-grade lesion), ASC-H, HSIL (high-grade lesion), up to carcinoma—plus glandular categories (AGC). It is the language in which a real Pap result is reported and acted upon.

TBS is the obvious candidate bridge, and we considered it seriously. It fails here for three reasons, and the third is the interesting one.

- **SIPaKMeD cannot populate TBS.** Its five classes are cell *types*, not diagnoses: *Parabasal*, *Metaplastic* and *Superficial-Intermediate* are normal cell categories, and *Koilocytotic* names an HPV cytopathic effect rather than a grade. The mapping to TBS is one-to-many at the fine level. Decisively, the dataset contains no high-grade category at all: a model trained on it *cannot express* HSIL—the grade that carries the clinical risk. A TBS axis with no HSIL is not a TBS axis.
- **The UCI target that would align with TBS carries no signal.** `Citology` is the natural counterpart of a Pap report. Under our leakage-safe protocol it scores AUROC = 0.488 (95% CI $[0.407, 0.575]$)—at or below chance. Aligning the constructs via TBS routes the system to the one target with nothing in it.
- **TBS does not touch the unit mismatch.** A Bethesda report describes a *slide*; SIPaKMeD describes a *cell*. Relabelling the axis does not tell you how to turn many cell verdicts into one patient verdict, and no public data can calibrate that aggregation because no patient identifiers exist.

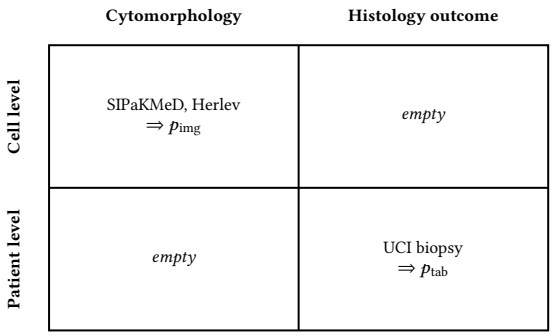


Figure 2. The mismatch, drawn. Our two experts occupy *opposite diagonal* cells: different construct *and* different unit of analysis. Calling their composition "P(abnormal)" is a definitional claim, not a calibration step.

What Bethesda actually teaches us

TBS would make the shared axis more clinically *legible* without making the two constructs *commensurable*—it relabels the problem rather than solving it. That is itself evidence for this proposal's central claim: the image label and the biopsy label are not two noisy readings of one variable. *Where a dataset natively provides TBS we use it*: HiCervix ships a three-level Bethesda hierarchy over 40,229 cells [3], and we use it for *external validation*—collapsing NILM to negative and LSIL/HSIL/ASC-US/ASC-H/carcinoma to positive, while *excluding* glandular categories the image expert provably cannot see.

5.3 The construct and unit mismatch

We therefore do *not* assert that p_{img} and p_{tab} measure the same event. We treat them as two *imperfect indicators of one latent state*, in the tradition of diagnostic-test evaluation without a gold standard [30], and we report every composed number *conditional on* that model, never as an unconditional probability of disease.

5.4 Problem setup and notation

Formal setup

Two frozen experts $f_k : x_k \mapsto p_k$, $k \in \{\text{img}, \text{tab}\}$, trained on cohorts D_{img} and D_{tab} with $D_{\text{img}} \cap D_{\text{tab}} = \emptyset$. No instance carries both modalities. $A \subseteq \{\text{img}, \text{tab}\}$, $A \neq \emptyset$, is the set of modalities actually available for a case. The task is a map from $\left(\{p_k\}_{k\in A}, \{\pi_k\}_{k\in A}, \tau\right)$ to a probability and a decision.

5.5 End-to-end architecture

5.6 Module cards

M1 — Image expert (borrowed).

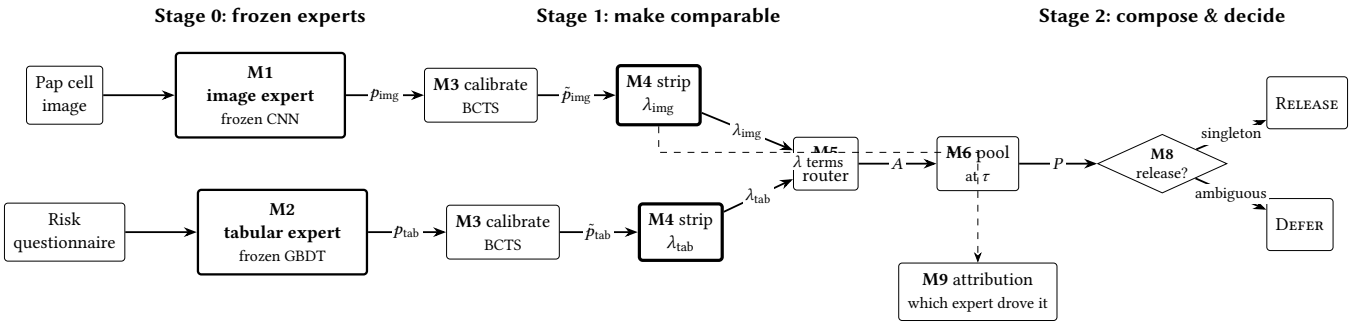


Figure 3. End-to-end architecture. Two experts trained on *disjoint* cohorts are frozen (Stage 0); each posterior is calibrated and then has its own training prevalence divided out (Stage 1); the router admits any non-empty subset and the pool applies one declared prior τ once (Stage 2). *Thick boxes are our contribution; thin boxes are deliberately borrowed.* Nothing after Stage 0 is trained on joint data, because no joint data exists.

Table 3. Notation.

Symbol	Meaning
$Y \in \{0,1\}$	latent target state (§5.2)
p_k	expert k's raw posterior
$\tilde{p}_k$	expert k's *calibrated* posterior
π_k	expert k's *training* prevalence
τ	*declared* deployment prevalence
λ_k	prior-stripped log-likelihood-ratio
w_k	reliability weight, $w_k \in [0,1]$
ρ	cross-modal dependence (non-identified)
σ	logistic function; logit $= \sigma^{-1}$

- *Purpose.* Emit a cytomorphology posterior for one cell.
- *Inputs / Outputs.* Cell image x_{img} / p_{img} and the training prevalence π_{img}.
- *Internal operations.* EfficientNet backbone [33], optionally distilled for edge use [18]; the five SIPaKMeD classes are collapsed to abnormal = {koilocytotic, dyskeratotic}.
- *Failure modes.* A 27–47 point external collapse (0.998 → 0.73/0.59/0.53); no patient IDs, so patient-wise splitting is impossible; koilocytosis marks *productive* HPV and is a weak proxy for high-grade disease.
- *Cost.* One forward pass.
- *Novelty.* None claimed.

M2 — Tabular expert (borrowed model, modified protocol).

- *Purpose.* Emit a pre-test risk from the questionnaire.
- *Inputs / Outputs.* x_{tab} / p_{tab}, $\pi_{tab} \approx 0.064$.
- *Internal operations.* Gradient-boosted trees [13] under a leakage-safe protocol: `Dx:*` dropped, resampling inside folds only, nested stratified CV, bootstrap CIs.
- *Failure modes.* 14.6:1 imbalance; 35–74 positives per target; a single Venezuelan cohort with no external validation anywhere in that literature.
- *Cost.* Seconds on CPU.
- *Novelty.* The *audit* is the contribution, not the model.

M3 — Per-expert calibration (borrowed).

- *Purpose.* Make each p_k a genuine frequency *before* any prior arithmetic.
- *Internal operations.* Bias-corrected temperature scaling with maximum-likelihood label-shift estimation [1], fitted on expert k's own held-out fold; report ECE, Brier and a reliability diagram per expert.
- *Why it is mandatory.* Prior correction presupposes a calibrated posterior—a condition "not true of modern neural networks" [1]. Skip M3 and M4 is meaningless.
- *Failure modes.* Few positives make the temperature high-variance; mitigated by cross-fitting.
- *Novelty.* None claimed.

M4 — Prior-stripping (our contribution). This is the module the proposal exists for, so it gets an analogy before the mechanism.

Two thermometers

Two nurses each estimate "is this patient feverish?". Nurse A works in a fever clinic where 40% of patients are feverish; Nurse B works in a general clinic where 6% are. Both are well-calibrated *in their own clinic.* If Nurse A says 70% and Nurse B says 70%, they are *not* saying the same thing: A's 70% is barely above her baseline of 40%, while B's 70% is a dramatic departure from her baseline of 6%. Averaging the two numbers silently treats the clinics' case mixes as if they were evidence about *this* patient. The fix is to ask each nurse not "how likely is fever?" but "*how much did this patient's signs move you away from your own baseline?*"—and to add the baseline back exactly once, at the end, using the prevalence of the clinic where the system is actually deployed.

- *Purpose.* Remove each expert's own cohort prevalence, so that what is pooled is *evidence* rather than evidence entangled with an accident of dataset construction.
- *Inputs / Outputs.* $(\tilde{p}_k, \pi_k)$ / a prior-free log-likelihood-ratio λ_k.
- *Internal operations.* Eq. (1); closed form, two logarithms.
- *Failure modes.* π_k must be *known*: for a curated cohort it is a construction artefact, not a population rate. We therefore treat π_k as a *declared, audited* quantity and sweep it ±50% (ablation A7). If $\tilde{p}_k$ is uncalibrated, λ_k is not a likelihood ratio at all.
- *Cost.* Microseconds; no training.
- *Novelty.* **New.** Closest prior work sits on either side and never crosses: one-classifier prior correction [1, 11, 31] versus common-prior combination [5, 10, 22].

M5 — Availability router (borrowed).

- *Purpose.* Accept any non-empty subset A.
- *Internal operations.* An absent expert contributes $\lambda_k = 0$, the additive identity—equivalently a vacuous opinion, the identity of Dempster's rule. *No imputation*, which at zero overlap has nothing to impute from.
- *Failure modes.* A lone weak expert must not auto-decide; it is routed to M8.
- *Novelty.* None claimed — this is ProbEn's marginalisation [5].

M6 — Composition at a declared prevalence (borrowed rule).

- *Purpose.* Produce the composed posterior with the prior counted exactly once.
- *Internal operations.* Eq. (2); the robust sum rule of Eq. (3) is evaluated alongside, since the tabular posterior is high-variance and the sum rule tolerates that better [22].
- *Failure modes.* The additive form *is* naive Bayes and inherits its over-confidence (M7).
- *Novelty.* None claimed. Only its *inputs* are new.

M7 — Dependence envelope (modified).

- *Purpose.* Quantify the price of the assumption we cannot avoid.
- *Internal operations.* Sweep $\rho \in [-1, 1]$ in Eq. (4) over the *real* marginals and report the range, not a point.
- *Failure modes.* It is a model-based *bound*, not a measurement, and is labelled as such on every figure.
- *Novelty.* Modified: the envelope is standard; transporting the biostatistical penalty [12] into a pre-registered prediction about our own output is the new part.

M8 — Per-path conformal abstention (borrowed, deliberately weakened).

- *Purpose.* A reject option with a distribution-free guarantee.
- *Internal operations.* Mondrian split conformal calibrated *per expert on its own cohort*; a non-singleton prediction set triggers Defer.
- *Honest scope.* We claim *marginal, per-path* coverage only. Per-expert guarantees do not compose into a joint one, and the label-shift correction requires unlabeled target data that does not exist here [27].
- *Novelty.* None claimed [2].

M9 — Modality attribution (modified).

- *Purpose.* Report *which* expert drove the call—free, because λ_{img} and λ_{tab} are on the same additive scale.
- *Failure modes.* Explanation is not benefit: clinician-facing explanations did not mitigate the harm of a biased model in a randomised study of 457 clinicians. We therefore claim no clinical benefit without a reader study.
- *Novelty.* Modest; a read-out of M4.

5.7 Mathematical formulation

Calibrate first (M3), then strip (M4):

$$\lambda_k \;=\; \operatorname{logit}\big(\tilde{p}_k\big) \;-\; \operatorname{logit}\big(\pi_k\big). \tag{1}$$

Reading Eq. (1) term by term: $\tilde{p}_k \in (0, 1)$ is expert k's calibrated probability that the case is abnormal; π_k is the fraction of abnormal cases in the cohort that expert was *trained* on. $\operatorname{logit}(x) = \log \frac{x}{1-x}$ converts a probability into log-odds, which turns the multiplication in Bayes' rule into addition. The subtraction is the whole idea: if the expert says exactly its own base rate ($\tilde{p}_k = \pi_k$) it has learned nothing about *this* case and $\lambda_k = 0$. The decisive quantity is the sign: $\lambda_k > 0$ means "this case looks *more* abnormal than my average case", $\lambda_k < 0$ means less. Because λ_k is a *likelihood ratio*, it no longer depends on the training prevalence—which is exactly why it is transportable to a different cohort, and why $\tilde{p}_k$ is not.

The composition applies one declared prior:

$$\operatorname{logit} P\big(Y{=}1 \mid \{x_k\}_{k\in A}\big) \;=\; \operatorname{logit}(\tau) \;+\; \sum_{k\in A} w_k \lambda_k. \tag{2}$$

Reading Eq. (2) term by term: τ is the prevalence of the population where the system is actually deployed—a *declared, auditable clinical input*, not something inherited from whichever dataset happened to be used. Each available expert adds its evidence λ_k, scaled by a reliability weight $w_k \in [0, 1]$. An absent expert contributes nothing ($\lambda_k = 0$), so $A = \{\text{img}\}$ recovers the image-only path *exactly*. With $w_k = 1$ this is precisely ProbEn / naive Bayes [5]; we claim no novelty in this line, only in what feeds it.

The robust alternative pools the anchored posteriors instead of the log-odds:

$$P_{\text{sum}} \;=\; \frac{1}{|A|} \sum_{k\in A} \sigma\big(\operatorname{logit}(\tau) + \lambda_k\big). \tag{3}$$

Algorithm 1 CO-PRIOR fit (per expert, one cohort at a time)

Require: cohort D_k, frozen expert f_k, target coverage $1-\alpha$
Ensure: (T_k, b_k), π_k, conformal thresholds $q_{k,c}$
1: split D_k into fit / calibration folds (stratified)
2: $\pi_k \leftarrow$ declared prevalence of D_k ▷ audited, not learned
3: $(T_k, b_k) \leftarrow \mathrm{BCTS}(f_k, \text{cal fold})$ ▷ M3
4: $\tilde{p} \leftarrow \sigma(z/T_k + b_k)$ on the cal fold
5: report ECE_k, Brier_k ▷ gate: M4 is invalid if uncalibrated
6: **for** $c \in \{0, 1\}$ **do**
7: $\quad q_{k,c} \leftarrow \mathrm{Quantile}_{1-\alpha}(\{1 - \tilde{p}_{i,c} : y_i = c\})$
8: **end for**
9: **return** $(T_k, b_k), \pi_k, q_{k,\cdot}$

Reading Eq. (3)*:* each expert is first turned back into a probability *at the deployment prevalence*, then averaged. Averaging cannot run away to 0 or 1 the way a sum of log-odds can, which is why Kittler et al. find the sum rule less sensitive to a single bad estimate [22]—relevant because our tabular expert is estimated from as few as 35 positives.

Finally the envelope over the unknown dependence:

$$\mathrm{AUROC}_{\mathrm{fused}}(\rho) \;=\; \Phi\left(\frac{d_{\mathrm{img}} + d_{\mathrm{tab}}}{\sqrt{2}\,\sqrt{2 + 2\rho}}\right), \quad d_k = \sqrt{2}\,\Phi^{-1}(\mathrm{AUROC}_k). \tag{4}$$

Reading Eq. (4) *term by term:* Φ is the standard normal CDF. d_k is the *separation* of expert k—how far apart its scores sit for abnormal versus normal cases—recovered from its measured AUROC. ρ is the correlation between the two experts' errors *given* the true state: $\rho = 0$ means they make independent mistakes (the optimistic case, where pooling helps most); $\rho = 1$ means they make the *same* mistakes, and the equation collapses to a single expert's AUROC—pooling adds nothing. Because both of our signals are driven by HPV, the honest operating region is $\rho > 0$. At our measured marginals (0.73 external image, 0.65 tabular) the gain is +0.016 at $\rho = 0$ and −0.030 at $\rho = 0.6$: composition stops paying almost immediately. The critical fact is that ρ *is non-identified at zero cohort overlap*—no estimator of it exists from our data—so we report the curve, not a number.

5.8 Algorithms

Reading Algorithm 1. Every line touches *one* cohort. Line 2 records the training prevalence as a declared quantity—we never estimate it from the other cohort, because the two cohorts share no patients. Line 3 fits the calibrator, and line 5 is a *gate*, not a report: if the expert is not calibrated, the prior-stripping in Eq. (1) does not yield a likelihood ratio and the rest of the pipeline is void. Lines 6–8 fit the conformal thresholds per class, on the same single cohort. Nothing here needs a paired sample—which is precisely why this proposal is testable.

Reading Algorithm 2. Line 1 seeds the accumulator with the *deployment* prior, so it is counted once and only once. The loop runs over *whatever arrived*: with only an image, only λ_{img} is added and the output is provably identical to the image-only path—graceful degradation is structural, not trained. Line 4 is the contribution. Line 8 asks the reject question against a per-expert conformal threshold, so the guarantee we quote is the one we can actually keep (marginal, per path). The returned $\{\lambda_k\}$ *is* the explanation (M9): the two numbers are on one scale, so "the image contributed +2.1, the questionnaire −0.3" is a complete account of the decision.

Algorithm 2 CO-PRIOR compose (inference, any subset)

Require: available $A \neq \emptyset$, raw $\{p_k\}_{k \in A}$, $\{(T_k, b_k), \pi_k, q_{k,\cdot}\}$, τ, weights w_k
Ensure: composed P, decision, attribution
1: $z \leftarrow \mathrm{logit}(\tau)$
2: **for** $k \in A$ **do**
3: $\quad \tilde{p}_k \leftarrow \sigma(\mathrm{logit}(p_k)/T_k + b_k)$ ▷ M3
4: $\quad \lambda_k \leftarrow \mathrm{logit}(\tilde{p}_k) - \mathrm{logit}(\pi_k)$ ▷ M4: strip
5: $\quad z \leftarrow z + w_k \lambda_k$ ▷ M6: pool
6: **end for**
7: $P \leftarrow \sigma(z)$
8: $C \leftarrow \{c : 1 - \tilde{p}_{k^*,c} \le q_{k^*,c}\}$ ▷ M8, k^* = strongest available
9: **if** $|C| \neq 1$ **then**
10: $\quad$ **return** P, Defer, $\{\lambda_k\}$
11: **end if**
12: **return** P, Release, $\{\lambda_k\}$

Table 4. Cost. Everything after the frozen experts is closed-form.

Stage	Fit	Inference
M1 image expert	hours, 1 GPU	one forward pass
M2 tabular expert	seconds, CPU	$O(1)$
M3 calibration	2 scalars/expert	$O(1)$
M4–M6	**none**	$O(\lvert A\rvert)$**, microseconds**
M8 conformal	one quantile/class	$O(1)$

5.9 Decision policy

Novelty boundary

The *prior-stripping precondition* is **our** contribution. The pooling rule, the router, the conformal reject option and the evidential conflict signal are deliberately **borrowed** and kept as separable modules—so the novelty stays cleanly the *operator*, not the pipeline around it.

5.10 Computational cost

The whole contribution runs on one 16 GB GPU because the only trained objects are two frozen experts and four scalars.

6 Novelty Statement

We do *not* claim the flexible any-modality architecture, the fusion rule, per-subset calibration, or modality dropout; those are [5], [10], [6] and [14] respectively, and we say so in the abstract. CO-PRIOR's differentiators, each checked against

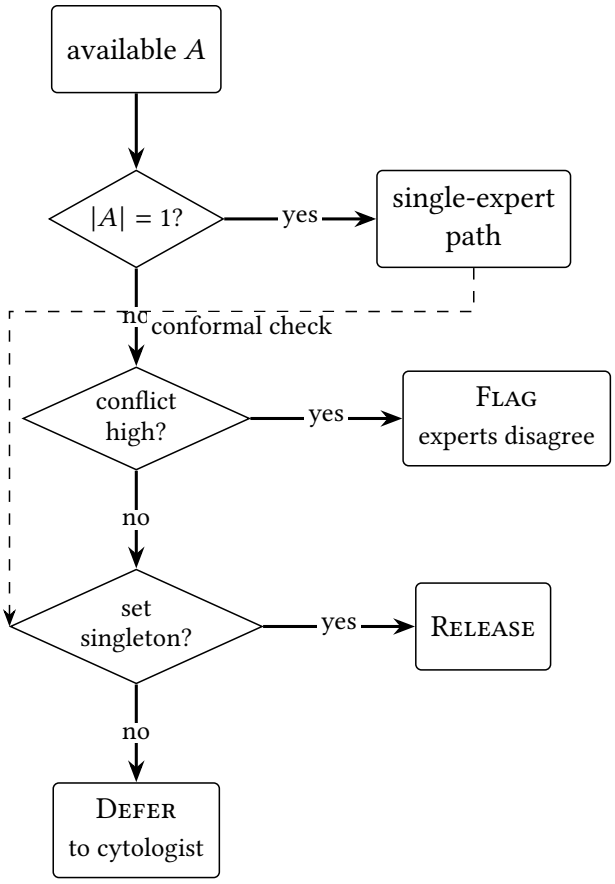


Figure 4. Decision policy. A lone weak expert never auto-releases; strong disagreement escalates rather than being averaged away.

the primary sources: **(1)** the *two-prior composition operator*—prior-stripping as a precondition of pooling, which the prior-correction literature (one classifier, one population [1, 11, 31]) and the combination literature (several classifiers, one common prior [5, 10, 22]) each leave to the other; **(2)** a *leakage-audited* honest baseline for the field's most-used risk dataset, which no prior paper supplies; and **(3)** a *pre-registered, directional* prediction of the over-confidence that a forced independence assumption must produce [12], reported as an envelope because the dependence is non-identified. The scope is equally explicit: the both-modality path is *unvalidatable* on public data, and we report that as a finding rather than papering over it with pseudo-pairing—a practice whose own dedicated critique we could locate [28] but not read.

7 Experimental Design

Model arsenal. Primary image expert: EfficientNet [33] (Apache-2.0, runs locally). Optional scale check: Smart-CCS [20], the only cervical foundation model with strong external numbers *and* a permissive licence (Apache-2.0, weights released). Tabular: gradient-boosted trees [13]. Note the licence inversion—general pathology foundation models are the *most* restricted, with some forbidding deployed use entirely, while the cervical-specific models are the most permissive. *Model-agnostic snapshot:* CO-PRIOR treats experts as black boxes emitting (p_k, π_k), so any of these can be swapped without touching M3–M9.

Table 5. Datasets and their roles.

Dataset	Role	What it tests
SIPaKMeD [26]	image train	in-distribution ceiling
Herlev	image external	transfer
HiCervix [3]	image external	Bethesda-graded transfer
UCI 383 [8]	tabular	honest ceiling, leakage
paired cohort	*none exists*	*the impossibility*

Table 6. Metrics. The headline is per-path net utility.

Metric	Definition / why
AUROC	discrimination; DeLong CI [7]
PR-AUC	mandatory at 4–12% prevalence
ECE / Brier	calibration; the headline safety metric
signed cal. error	*direction* of miscalibration (H-E)
coverage	conformal, per path, marginal
deferral rate	vs plausible cytologist workload

Baselines. Image-only (external, *not* in-distribution); tabular-only (leakage-safe); **`max(image-only, tabular-only)` — the key head-to-head**; raw pooling without stripping (isolates M4); ProbEn with a common assumed prior [5]; the sum rule [22]; and a deliberate *leaky reproduction* (SMOTE-before-split + `Dx:*`) to show the published 0.98–1.00 is an artefact.

Pre-registered targets

Every number in Table 7 is a bar fixed *before* the experiments are run. The falsifier row is included deliberately: we expect it may fire.

Statistical analysis.

- AUROC differences: DeLong for correlated ROCs on the same cases [7]; BCa bootstrap (10,000 resamples) elsewhere.
- ECE differences: paired bootstrap with BCa intervals.
- Multiplicity: the primary family is {H-C, H-E}; Holm–Bonferroni. Everything else is secondary and labelled so.

Table 7. Pre-registered acceptance criteria.

Claim	Metric	Numeric bar (pre-registered)	Test	Result artifact
H-A honest ceiling	AUROC	upper 95% CI ≤ 0.75	BCa bootstrap, 10,000	CI table + forest plot
H-B leakage explains it	ΔAUROC	honest + leaks reconstructs published within 0.02	two-factor ablation	grouped bar + 95% CI
H-C stripping helps	ΔECE	< 0, CI excludes 0	paired bootstrap	reliability diagrams
H-D confidence monotone	mean conf.	0 superset violations	bootstrap over patterns	confidence-vs-subset plot
H-E falsifier	AUROC	fused $\leq$ `max(single)` at $\rho \geq 0.3$	DeLong [7]	envelope curve

Table 8. Experiment grid.

ID	Experiment	Data	Comparison	Artifact
E1	leakage-safe UCI re-benchmark	UCI	vs published 0.98–1.00	CI table
E2	leakage decomposition	UCI	`±Dx:*` $\times$ ±pre-split resample	grouped bar chart
E3	per-expert calibration	UCI, SIPaKMeD	none/temp/BCTS/isotonic	reliability diagrams
E4	**prior-stripping**	both	stripped vs raw vs post-hoc	ECE bar + CI
E5	dependence envelope	real marginals	fused vs `max(single)`	envelope curve
E6	external transfer	Herlev, HiCervix	in-dist vs external	transfer matrix
E7	any-subset degradation	both	{img}, {tab}, {both}	confidence-vs-subset plot
E8	abstention	both	risk–coverage	AURC curve

Table 9. Ablation grid. A6 is the ablation the proposal lives or dies by.

ID	Removed / changed	Claim it isolates	Expected effect (grounded)	Artifact
A3	`Dx:*` back in	target leakage (G5)	0.592 $\rightarrow$ 0.998 (measured)	bar chart
A4	resample before split	resampling leakage	reproduces $\approx$ 0.98–1.00 [29]	bar chart
A5	target choice	estimand	Citology 0.488, Hinselmann 0.450 are $\leq$ chance	CI table
A6	**stripping OFF**	**M4 — the contribution**	**if ΔECE ≈ 0 the claim dies**	ECE bar + CI
A7	π_k mis-specified ±50%	robustness of M4	degradation grows with error	line plot
A8	τ sweep	declared prior	the fused number is meaningless without τ	line plot
A9	pool rule	product vs sum	sum more robust at 35 positives [22]	bar chart
A11	calibration off	M3 is a precondition	λ ceases to be an LLR	reliability diagram
A14	evidential u as weight	G6	degrades: u is relative, not absolute	bar chart

- Variance: ≥ 5 repeats of stratified CV with recorded seeds; report mean ± SD *and* the pooled-OOF bootstrap CI.
- *Power, stated honestly:* with 55 biopsy positives the AUROC CI half-width is ≈ 0.07. We therefore pre-declare that tabular-arm differences below 0.05 AUROC are **not resolvable** and will not be claimed.

Result artifacts. Each analysis has one required output: reliability diagrams (E3, E4, A11); grouped bar + 95% CI (E2, A3, A4); the dependence envelope curve (E5, H-E); a transfer matrix heatmap (E6); a confidence-vs-availability plot (E7, H-D); a risk–coverage (AURC) curve (E8); and a CI/forest table (E1, A5).

8 Threats to Validity and Limitations

- *Construct (severe).* p_{img} and p_{tab} measure different things on different units (Fig. 2). *Mitigation:* latent-indicator framing [30]; every composed number is reported conditional on that model.
- *Internal.* π_k for a curated cohort is a construction artefact. *Mitigation:* declared and audited, swept ±50% (A7).
- *External (fatal to the headline path).* No paired cohort exists, so joint discrimination and calibration are unmeasurable. *Mitigation:* none possible; we report it as the finding and cost a prospective pilot instead.
- *Statistical.* 35–74 positives per UCI target; differences < 0.05 AUROC are not resolvable and are not claimed.

- *Clinical.* A weak questionnaire must never lower a positive cytology call. *Mitigation:* the questionnaire may only raise suspicion or trigger deferral.
- *Novelty risk.* If A6 shows prior-stripping is numerically equivalent to a single post-hoc correction, the operator is a re-parameterisation and the contribution collapses—in which case this work is demoted to a *diagnostic study*: the leakage audit plus the impossibility result, reported as such.

9 Conclusion

A flexible cervical screener that accepts an image, a questionnaire, or both is blocked not by architecture but by data: the two public signals share no patients, no labels, and no unit of analysis. We proposed CO-PRIOR, whose single claim is small and testable—strip each expert's own training prevalence out before pooling, and apply one declared deployment prior afterwards—because that is the one operation the prior-correction and classifier-combination literatures each leave to the other, and because it can be validated one cohort at a time, which is the only validation this data permits. Everything else is deliberately borrowed and named. The leakage-audited baseline (0.656 against a published 0.98–1.00) and the pre-registered falsifier are what make the proposal falsifiable rather than merely plausible; if fusion does not beat the better single expert, we will say so.

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